

Minseok Kim, M.S. Student

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Research Interests

📌 **Goal:** Learning-based control for energy-efficient, human-like bipedal locomotion.

📌 **Research Topics:**

- Reinforcement learning-based control for robotic systems.
- Model-based control approaches for dynamic systems.

Education

- 2024/09 – present 📌 **M.S., Mechanical and Robotics Engineering**
Gwangju Institute of Science and Technology (GIST), Gwangju, Korea
- 2019/03 – 2024/08 📌 **B.S., Mechanical System Engineering** (Graduated early)
Jeonbuk National University, Jeonju, Korea.

Experience

- 2025/07 📌 **Completed AI & Robotics Summer School 2025**
Vision-based object detection and manipulator control with Isaac Lab.
- 2025/03 📌 **Teaching Assistant**, Signals and Systems
School of Mechanical and Robotics Engineering, GIST
- 2025/02 📌 **Teaching Assistant**, Samsung Motor Bootcamp
- 2024/09 📌 **Teaching Assistant**, Dynamics
School of Mechanical and Robotics Engineering, GIST
- 2024/08 📌 **Photointerrupter-based Latching Switch Circuit Design**
Developed latching switch circuit using photointerrupters and relays.
- 2024/03 📌 **Capstone Design**
Motor control system to prevent food mixer pinching accidents.
- 2023 📌 **Undergraduate Research Student**

Projects

- 2025 – Present

Bipedal Robot Control
 - End-to-end control of a bipedal robot using deep reinforcement learning.
 - Developing a control policy using PPO in Isaac Gym for stable and human-like bipedal locomotion in simulation.
- 2024/12

Nonlinear Control (Course Project)
 - : Evaluation of Local Stabilization of an Inverted Pendulum on a Cart Under Perturbations.

Awards

- 2024/06

2024 Capstone Design **Bronze Award**
- 2024/02

2023 University President’s Award (**Academic Excellence Award**)

Skills

- Languages

Korean, English
- Coding

Python, C/C++
- CAD Tools

Fusion 360, CATIA, KiCad
- Simulation & Robotics

Isaac Gym, Isaac Sim, MuJoCo
- Embedded System

Arduino, Raspberry Pi, Teensy 4.1

Extra-Curricular Activities

- 2024/01 – 2024/02

Study abroad
Asia Pacific University | Malaysia
- 2020/03 – 2021/09

Military Service
Republic of Korea Army (ROK Army)

References

Available on Request